CARTER GRIMMEISEN

SOFTWARE ENGINEER

CONTACT

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EDUCATION

MS COMPUTER SCIENCE
Expected 2027 - 4.0 GPA
University of Alabama in Huntsville

BS COMPUTER SCIENCE
May 2022 - 4.0 GPA
University of Alabama in Huntsville

SKILLS

LANGUAGES
Python, TypeScript, Ada, Java, C++, Bash, Rust, C#, Kotlin, C, HTML/CSS/JS

FRAMEWORKS AND TOOLS
React Native, Flutter, Vue.js, SvelteKit, Node.js, PyTorch, PostgreSQL, SQLite/DuckDB, Docker, CI/CD, Git, Express.js, Flask, Plotly Dash, MPI, React, Svelte, Electron, GitLab, GitHub, Docker Compose, MongoDB, Conda, Cython

Active DoD Secret clearance

SUMMARY

Software engineer with 5 years of experience, 4 as a tech lead. Delivers greenfield products end-to-end across web, mobile, ML, and DevOps: scopes requirements with customers, writes the technical approach, and mentors engineers.

EXPERIENCE

PRINCIPAL SOFTWARE ENGINEER — MAY 2026 - PRESENT

TECHNICAL CONSULTING SOLUTIONS
Subcontractor to Northrop Grumman
+ Develop on a large, fielded legacy Ada codebase on a 10-12 person team
+ Built a web-based log analysis tool that processes log files as they reach the log server; developers and testers use it to explain system behavior, and it exposed a class of slowdowns behind timeouts and failures
+ Own vulnerability scanning and metrics reporting

LEAD SOFTWARE ENGINEER — JUNE 2022 - MAY 2026

CFD RESEARCH CORP.

LEADERSHIP
+ Led a 3-person team in an 8-10 person group: performance reviews, mentoring, hiring interviews, and matching people to projects
+ Ran greenfield SBIR projects ($800K-$6M) end-to-end for scientist teams, from requirements to customer reviews, across mobile, web, ML, and DevOps
+ Wrote the technical approach for ~6 proposals: 3 won outright and 1 advanced jointly; the expertise supported a separate ~$10M win
+ Contained scope creep with workarounds and phased delivery, and brought alternatives when approaches failed

DELIVERY
+ Cut ML training from ~8 hours to under 30 minutes (20x) with a SQLite/DuckDB-backed PyTorch dataloader
+ Built a simulation dashboard for ~12 scientists, replacing a manual script-and-CSV workflow: Postgres job queue, custom scheduler, and autonomous CI/CD
+ Led development on an offline-first React Native app with 2D/3D visualization (VTK.js) for blast overpressure exposure, used by soldiers and trainers on dozens of devices
+ Led a team of scientists and developers in parallelizing, packaging, and documenting a space-weather simulation toolkit for HPC deployment
+ Built the real-time monitoring and configuration web UI (SvelteKit) for a spin-out's software-defined firewall product
+ Built a Python application packager with a custom Python distribution builder, handling licensing, compilation, and packaging to streamline deployment of internal apps and libraries (Python, Docker)
+ Delivered 3D visualization and web scraping work for scientist teams

SOFTWARE ENGINEER, INTERN — MAY 2020 - MAY 2022

NORTHROP GRUMMAN
+ Automated an ~18-hour biweekly vulnerability-scan reporting process, bringing recurring engineering time to zero
+ Built an Ada mocking framework for unit testing a legacy codebase, on a 7-8 person DevOps team
+ Managed builds and the CI/CD pipeline for the team

SOFTWARE DEVELOPER — MARCH 2017 - MAY 2020

AVILUTION, LLC.

+ One of two engineers at an early-stage avionics startup (now dozens of employees); owned architecture, deployment, and support
+ Built a drag-and-drop display designer (Electron, Vue.js, TypeScript) with dozens of widgets, plus aircraft configuration management; a core selling point
+ Implemented ECDSA signing of flight-critical configurations and contributed to core avionics C code
+ Built low-level hardware drivers and a custom embedded Yocto build as part of an avionics-as-software stack

OPEN SOURCE

ZOD-PRISMA — https://www.npmjs.com/package/zod-prisma

MADE WITH TYPESCRIPT, PRISMA, ZOD

Generates Zod validators from Prisma schemas. **900+ GitHub stars**, **2.3M lifetime downloads**.